

RENANCE

OSUN STATE UNIVERSITY, OSOGBO

NSC 421 — Past Questions & Materials

by Renance

Course: NSC 421

Institution: Osun State University, Osogbo

Papers in this pack: 1

Levels: 500

This study pack was compiled and restructured by Renance for NSC 421 students at Osun State University, Osogbo. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — MATERNAL AND CHILD HEALTH NURSING AND NORMAL MIDWIFERY (NSC 421/422)

Level 500 • First Semester • Handout

MATERNAL AND CHILD HEALTH NURSING
AND NORMAL MIDWIFERY
(NSC 421/422)
(3 Units)

S. J Ayamolowo

Department of Nursing Science

Faculty of Basic Medical Sciences

Obafemi Awolowo University, Nigeria

TOPIC:

- Infertility and Assisted Reproductive Technology

Objectives

- At the end of this lecture, you should be able to:
 - Give different definitions of infertility
 - Explain the time trends and geographic variations in infertility
 - Identify the etiology and discuss the treatment of male and female factor infertility
 - Explain the adverse effects of infertility treatment

Introduction

- Infertility is defined as a failure to conceive within one or more years of regular unprotected intercourse.
 - >2 years (WHO)
- Infertility means not being able to get pregnant after one year of trying. Or, six months, if a woman is 35 or older.
- Women who can get pregnant but are unable to stay pregnant may also be infertile
- Infecundity: Inability to achieve a live

Introduction

- Pregnancy is the result of a process that has many steps
 - To get pregnant: A woman must release an egg from one of her ovaries (ovulation).

- The egg must go through a Fallopian tube toward the uterus (womb).
- A man's sperm must join with (fertilize) the egg along the way.
- The fertilized egg must attach to the inside of the uterus (implantation).

Types of Infertility

- Primary infertility: refers to couples who have not become pregnant after at least 1 year of unprotected sex (intercourse).
- Secondary infertility: refers to couples who have been pregnant at least once, but never again.
- Demographic Definitions:
 - Primary infertility: Absence of a live birth at specific ages (e.g. > age 30) in non-contracepting population
 - Secondary infertility: Absence of a live birth > 5 years in persons with prior births

Causes of Infertility: Female

- Common causes of infertility of females include:
 - Ovulatory disorders
 - Tubal occlusion due to Pelvic Inflammatory

Diseases (PID)

- Age-related factors: older age
- Uterine problems
- Previous tubal ligation
- Endometriosis: scarring from sexually transmitted infection or endometriosis

Causes of Infertility: Female

- Cancer or tumor: Growths (such as fibroids or polyps) in the uterus and cervix
- Clotting disorders, Diabetes, Thyroid disease
- Ovarian cysts and polycystic ovary syndrome (PCOS)
- Birth defects that affect the reproductive tract
- Excessive exercising, Eating disorders or poor nutrition
- Use of certain medications, including chemotherapy drugs
- Drinking too much alcohol, Obesity
- Too little or too much hormone.

Causes of Infertility: Male

- Male infertility may be due to:

- A decrease in sperm count, Sperm being blocked from being released, Sperm that do not work properly
- Heavy alcohol use, drugs, environmental toxins, including pesticides and lead, smoking cigarettes
- Health problems such as mumps, serious conditions like kidney disease, or hormone problems

Causes of Infertility: Male

- Use of certain drugs, such as cimetidine, spironolactone, and nitrofurantoin
- Radiation treatment and chemotherapy for cancer
- Being in high heat for prolonged periods
- Birth defects , Too little or too much hormones
- Impotence, Infection, Older age, Scarring from sexually transmitted diseases, injury, or surgery
- Retrograde ejaculation

Investigation of Infertility

- Initial Visit: It is the most important
- The infertility is a problem of couple
- The male partner should be present
- History: both male and female to guide diagnostic and treatment plans
- History: medical, Surgical, Menstrual, Previous obstetrical, Contraceptive, Sexual problems

Investigation of Infertility: Women

- Diagnostic Tests, Blood and imaging tests will be done.
- Blood tests to check hormone levels, including progesterone and follicle stimulating hormone,
- Checking body temperature first thing in the morning to check if the ovaries are releasing eggs
- Hystero-salpingography (HSG)(X-ray of the uterus and Fallopian tubes),
- Pelvic ultrasound, Laparoscopy,
- Luteinizing hormone urine test (ovulation prediction),
- Thyroid function tests

Investigation of Infertility: men

- SEMEN ANALYSIS: A normal semen analysis show: semen volume:1.5 to 5 ml, pH - 7.2 -7.8, concentration:>20 million

sperm/ml, total sperm count: >50 to 60 million per ejaculate,

Motility >50 %, morphology >30 % normal.

- Any abnormality in this result will show chances of infertility.
- POSTCOITAL TEST # SPERM PENETRATION TEST # TESTICULAR BIOPSY # IMMUNOLOGICAL TESTS # CHROMOSOMAL STUDIES
- Male factor infertility account for about half of all infertility problems.
- Problems for the male can include:
 - Lack of Sperm - AZOOSPERMIA
 - Abnormal Sperm- TERATOSPERMIA
 - Absence of semen-ASPERMIA
 - Impaired motility of sperm- ASTHENOSPERMIA
 - Dead sperms-NECROSPERMIA

Treatment of Male Infertility

- Sexual problems: help men deal with impotence or premature ejaculation.
- Behavioral therapy and/or medicines can be used in these cases.
- Too few sperm: Sometimes surgery can correct the cause of the problem.
- In other cases, surgically remove sperm directly from the male reproductive tract..

Treatment of Male Infertility

- Antibiotics can also be used to clear up infections affecting sperm count.
- Sperm movement: Sometimes semen has no sperm because of a block in the man's system.
- In some cases, surgery can correct the problem.
- Testosterone: It improves testicular function.
- Testosterone patch is applied directly to the scrotum (TESTODERM) OR ANDRODERM.
- Sildenafil (Viagra): It improves ERECTILE DYSFUNCTION

Treatment of Female Infertility

□Tubal Occlusion

- Assisted Reproductive Technologies (ART)
- In vitro fertilization (IVF)
- Gamete intrafallopian transfer (GIFT)
- Zygote intrafallopian transfer (ZIFT)
- Tubal surgery

Treatment of Female Infertility

□Ovulation Disorders: Ovulation induction:

- 1. Clomiphen □ ER binding GnRH,FSH/LH dosage □ 50 mg, period day 5th □ 5days □
- 2. Gonadotropin therapy □ Indications: Hypogonadotropic hypogonadism; Pituitary dysfunction; COH (controlled ovarian hyperstimulation) in IVF; HMG: human menopausal gonadotropins; FSH 75 IU/LH 75 IU, IM or SC; Recombinant FSH: 75 IU, SC;
- 3. HCG: 5000-10000 IU;
- 4. Gonadotropin releasing hormone agonist (GnRH- □) □ hypothalamic factor, as COH; e.g zoladex; decapeptyl, dipherenline, enantone;
- 5. GnRH antagonist: usually injected or given with a nasal spray
- 6. Bromocriptine :(Parlodel): used for women with ovulation problems due to high levels of prolactin.

Treatment of Female Infertility

□Endometriosis

- Drug treatment (Danazol)
- Surgery

Assisted Reproductive Technology (Art)

- Success rates vary and depend on many factors.
- Success rate of ART include may be affected by:
 - age of the partners,
 - reason for infertility,
 - clinic, type of ART,
 - if the egg is fresh or frozen,
 - if the embryo is fresh or frozen

Common Methods of ART

- Intrauterine insemination (IUI)
- In vitro fertilization (IVF) .
- Zygote intra fallopian transfer (ZIFT) or Tubal

Embryo Transfer is similar to IVF.

- Gamete intra fallopian transfer (GIFT)
- Intra cytoplasmic sperm injection (ICSI)
- Surrogacy
- Gestational Carrier

Intrauterine insemination (IUI)

- Often called artificial insemination.
- The woman is injected with specially prepared sperm.
- Sometimes the woman is also treated with medicines that stimulate ovulation before IUI.

- Indications:
 - Mild male factor infertility
 - Women who have problems with their cervical mucus
 - Couples with unexplained infertility
- Types:
 - 1. artificial insemination with husband's sperm (AIH);
 - 2. artificial insemination by donor (AID);
- Method: placement of about 0.3 ml of washed, processed and concentrated sperm into the intrauterine cavity by trans-cervical catheterization

In Vitro Fertilization and Embryo Transfer (IVF-ET)

- Fertilization outside of the body.
- IVF is the most effective ART.
- Indications: 1. tubal factor; 2. endometriosis; 3. unexplained infertility; 4. IUI failure; 5. Immunologic factors;
- Method:
 - 1. Superovulation: COH, GnRH- α FSH(HMG)/HCG;
 - 2. Aspiration of eggs;
 - 3. Fertilization with capacitated sperm;
 - 4. Culture of fertilized egg in the lab;
 - 5. Replacement of fertilized egg into the uterus (After 3 to 5 days)

In Vitro Fertilization and Embryo Transfer (IVF-ET)

- Risks:
 - Requires a significant physical, emotional, financial, and time commitment.
 - Stress and depression are common among couples dealing with infertility.
 - Fertility drugs may cause ovarian hyperstimulation syndrome (ohss), decreased urination despite drinking plenty of fluids, nausea, vomiting, and shortness of breath.
- Complications:
 - Multiple Births, Birth defects, Pre-eclampsia, Ovarian hyperstimulation syndrome (OHSS) Premature birth, Low birth weight, Long term emotional, social and psychological impact

In Vitro Fertilization and Embryo Transfer (IVF-ET)

- Aftercare:
 - After the IVF procedure is performed the woman can

resume normal activities.

- A pregnancy test can be done approximately 12-14 days later to determine if the procedure was successful.
- Daily shots or pills of the hormone progesterone for 8 - 10 weeks after the embryo transfer.
- Progesterone is a hormone produced naturally by the ovaries that helps thicken the lining of the womb (uterus).
- Too little progesterone during the early weeks of pregnancy may result in a miscarriage.

Zygote Intra Fallopian Transfer (ZIFT)

- Also called Tubal Embryo Transfer
- It is similar to IVF.
- Fertilization occurs in the laboratory.
- Then the very young embryo is transferred to the Fallopian tube instead of the uterus.

Gamete Intra Fallopian Transfer (GIFT)

- Involves transferring eggs and sperm into the woman's Fallopian tube.
- So fertilization occurs in the woman's body.
- Few practices offer GIFT as an option.
- Indications:
 - 1. unexplained infertility; 2. endometriosis;
 - 3. IUI failure; 4. Premature ovarian failure (POF);
 - 5. Immunologic factors;
- Method: 1. Superovulation is induced as IVF-ET; 2. HCG injection is given; 3. Follicle are aspirated via laparoscopy; 4. Sperm mixed with egg; 5. Placement of Sperm mixed with egg into fallopian tube.

Intra Cytoplasmic Sperm Injection (ICSI)

- It is often used for couples in which there are serious problems with the sperm.
- Sometimes it is also used for older couples or for those with failed IVF attempts.
- In ICSI, a single sperm is injected into a mature egg.
- Then the embryo is transferred to the uterus or Fallopian tube.

Other ART Procedures

- TUBAL REVERSAL: individuals with previously tubal ligation can undergo a microscopic tubal

ligation reversal or an In Vitro Fertilization (IVF) procedure.

- A tubal reversal is the better option for most patients.
- Success rate with microsurgical tubal reversal is about 95% when surgeon has good skill.
- Needs one small incision and 1 day hospitalization.

Other ART Procedures

- MICROSCOPIC VASECTOMY REVERSAL: allows men who have previously undergone vasectomy to become fertile again.
- The vas deferens is microsurgically reconnected and the epididymal blowouts are microsurgically bypassed, allowing sperm to travel out of the epididymis and into the EJACULATION
- MICRO ASSISTED FERTILIZATION(MAF): It occurs in vitro and is expensive
- Used in IVF or GIFT fails, Immunological derived infertility, Inability of sperm to penetrate egg.

Role of Nurse In Infertility Management

- ASSESSMENT:
 - The nurses role during this stage is to educate the couple about each test and investigation, including why and how this investigation to be performed.
 - The nurse plays a vital role in alleviating the fear and anxiety about the various diagnostic procedure.
- TREATMENT:
 - The nurse plays the link between the doctor and couple and should always be available to couple for their assistance, guidance and support before, during and after infertility treatment.
 - Numerous ethical issues are associated with infertility treatments and the couple undergoing treatment need appropriate counseling and discussion.

Role of Nurse In Infertility Management

- EDUCATION:
 - The role of nurse in educating the patients includes education about the basic male and female anatomy and physiology and how the drug acts on their body, including possible side effects.
 - Fertility nurse should also educate the couple about self-

administer medications.

- PREVENTION OF INFERTILITY :

- Avoid gonadotoxins
- Decrease exposure to occupational and environmental hazards
- Avoid transmission of STDs by limiting the number of sexual partners and by using condoms
- Eating a well-balanced and nutritious diet.
- Stopping smoking and drinking.

Role of Nurse In Infertility Management

- PSYCHOLOGICAL SUPPORT:

- A couple undergoing infertility treatment are usually stress due to a variety of reason.
- Because infertility evaluation and treatments are expensive, time consuming, invasive, stressful and not always successful and also not socially acceptable.
- Proper emotional support and guidance is required by the couple at this stage.
- The nurse as a counselor should provide anticipatory advice and guidance about the normal range of expectations and reasons throughout the treatment.

Assignment

- Discuss briefly the following alternatives to childbirth i.e as possible solutions to for subfertility:

1. Surrogate motherhood
2. Adoption
3. Child-free living

Reference

- Fraser, D.M., Cooper, M.A and Nolte, A. G. W. (eds.) 2006. Myles Textbook for Midwives. African Edition. Churchill Livingstone. Pp 103-111.
- Pillitteri, A. (2014). Maternal and child health nursing: care of the childbearing and childrearing family. 7th ed. Philadelphia: Lippincott Williams & Wilkins
- The Johns Hopkins University and Ronald Gray(2007): Female and male infertility. Johns Hopkins University